The canonical-coloring theorem

Mathematical assessment and working guide for Alex Chan
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1. Assessment and exact scope

Assessment: I find no mathematical gap in the uploaded all-orders argument. With the classical Burnside–Schur theorem as an external mathematical input, it proves the prescribed-coloring conjecture for *every* cyclic tournament. The mathematical review is supplemented by a completed Lean proof, but no external human peer review is claimed. The Lean formalization was completed on 30 October 2026; its final theorem axiom reports contain only Lean's three standard logical axioms (Section 5). A bounded literature search did not locate a later complete resolution; priority is not established.

Write $n = 2q + 1 \geq 3$, and let S contain exactly one of $d, -d$ for each nonzero residue modulo n. The tournament $T(n; S)$ has an arc $i \to j$ when $j - i \in S$. The required coloring is

$$\lambda(i) = 1 \ (0 \leq i \leq q), \qquad \lambda(i) = 2 \ (q < i \leq 2q).$$

It is *distinguishing* when its only color-preserving automorphism is the identity. It need not be a proper vertex coloring.

This is exactly Conjecture 2 in Meslem–Sopena [1]. The original Albertson–Collins note [2, Conjecture 2.4] uses the smaller interval $L = \{0, \ldots, q - 1\}$. Its complement is $\{q, \ldots, 2q\} = I_n + q$, where $I_n = \{0, \ldots, q\}$. A set and its complement have the same stabilizer, and translating a set conjugates its stabilizer. Thus these formulations are equivalent. The existence of *some* distinguishing two-coloring was already known from Gluck's theorem; the prescribed interval is the result at issue.

Stronger theorem. If n is a positive odd integer and $G \leq \mathrm{Sym}(\mathbb{Z}_n)$ has odd order and contains $\tau : x \mapsto x + 1$, then the only $g \in G$ satisfying $g(I_n) = I_n$ is the identity.

A *regular subset* means a subset with trivial setwise stabilizer. The theorem covers all connector sets without using their details. There is an infinite family with 2^q connector sets at order $2q + 1$, so no finite collection of graphs alone can prove the statement.

External input. Burnside–Schur [3, opening theorem, p. 1] says that a primitive action with a cyclic regular subgroup is either doubly transitive or has prime degree and is conjugate to an affine action over the prime field. The action-level conclusion is essential: an abstract group isomorphism would not suffice. In the completed formalization, this input is supplied by imported proof terms from AxiomMath/CFSG, not an unproved axiom or theorem parameter.

What changed in the new paper. The supplied induction and coordinate-normalization argument are retained. An elementary interval-intersection lemma replaces the finite-field second-moment calculation. Both versions have been checked.

2. Full proof: the block-system step

We prove the stronger theorem by strong induction on odd n. For $n = 1$, there is only the identity permutation. Assume the claim is known for every smaller positive odd degree.

First record two consequences of the induction hypothesis. A set and its complement have identical stabilizers. Also, if a group contains all translations, then regularity of I implies regularity of $I + t$: a permutation h preserving $I + t$ gives $\tau^{-t}h\tau^{t}$ preserving I. These facts handle the smaller half-intervals that appear inside blocks.

The group G is transitive because it contains the full n-cycle τ. Suppose it has a nontrivial invariant block system. Let B be its block containing 0. For $t \in B$, the point t belongs to both B and $B + t$. The block property gives $B + t = B$. This implies closure under addition and also gives $-t \in B$, since $0 \in B + t$. Hence B is an additive subgroup of $\mathbb{Z}_n$.

Every subgroup of $\mathbb{Z}_n$ consists of multiples of a divisor a of n. Let $n = ab$. All blocks are translates of B, so they are

$$B_r = \{r + at : 0 \le t < b\}, \qquad r = 0, \dots, a - 1.$$

Nontriviality gives $1 < a, b < n$, and both factors are odd. Write $a = 2u + 1$ and $b = 2v + 1$. Since $(ab - 1)/2 = av + u$,

$$I_n \cap B_r = \begin{cases} \{r + at : 0 \le t \le v\}, & r \le u, \\ \{r + at : 0 \le t < v\}, & r > u. \end{cases}$$

Thus the intersection counts are $v + 1$ exactly for block indices $0, \dots, u$, and v for the others. These coordinates have $0 \le r + at < ab$, so there is no wraparound in the calculation.

Take $g \in G$ with $g(I_n) = I_n$. Its induced permutation on blocks preserves these counts, hence preserves I_a. The full induced group is an image of G, so has odd order. It contains $r \mapsto r + 1$ on $\mathbb{Z}_a$. Induction makes the induced permutation of g the identity. Consequently g fixes each block *setwise*.

Now restrict to B_r. The image on B_r of its setwise stabilizer in G has odd order. It contains τ^a, which is $t \mapsto t + 1$ in the coordinate $r + at \leftrightarrow t \in \mathbb{Z}_b$. The preserved subset is either $I_b = \{0, \dots, v\}$ or $\{0, \dots, v - 1\}$. The latter has complement $\{v, \dots, 2v\} = I_b + v$. Induction, complementing and translating as appropriate, therefore makes the restriction of g the identity. This holds in every block, so g is the identity.

Example at $n = 9$. For $a = b = 3$, the blocks are $\{0, 3, 6\}, \{1, 4, 7\}, \{2, 5, 8\}$. Their intersections with $I_9 = \{0, 1, 2, 3, 4\}$ have sizes $2, 2, 1$. The quotient argument fixes the blocks; the within-block argument fixes their points.

3. Full proof: the primitive step and the tournament

Suppose the action is primitive. It cannot be doubly transitive: such a group has order divisible by the number $n(n-1)$ of ordered distinct pairs, which is even for $n > 1$ odd. Burnside–Schur therefore gives prime degree $n = p$ and a bijection $f : \mathbb{Z}_p \to \mathbb{F}_p$ conjugating G into $\mathrm{AGL}_1(p)$.

The original labels are retained. The permutation $f\tau f^{-1}$ has order p. An affine map $z \mapsto \alpha z + \beta$ with $\alpha \neq 1$ has a fixed point; moving that point to 0 conjugates it to multiplication by α. Its order divides $p-1$. Hence $f\tau f^{-1}$ is translation by some $c \neq 0$, and

$$f(j+1) = f(j) + c, \qquad f(j) = f(0) + jc.$$

Renormalizing by $z \mapsto (z - f(0))/c$ gives the original residue coordinate. Every element of G is therefore affine *in the coordinates defining* I_p.

Affine interval lemma. Let $p = 2q+1$ and $I = \{0, \dots, q\}$. For a residue d put $C(d) = |I \cap (I+d)|$. For $1 \le d \le q$, $I \cap (I+d) = \{d, \dots, q\}$; also $C(-d) = C(d)$ by translation. Thus

$$C(0) = q+1, \qquad C(d) = q + 1 - \min\{d, p-d\} \quad (1 \le d \le 2q).$$

The largest value at nonzero shifts is q, occurring exactly at $d = \pm 1$. If $h(x) = \alpha x + \beta$ preserves I, then it maps $I + d$ to $I + \alpha d$, giving $C(\alpha d) = C(d)$. Taking $d = 1$ forces $\alpha = \pm 1$.

If $\alpha = 1$, a nonzero β would imply $C(\beta) = |I| = q+1$, impossible. Thus h is the identity. The reflection $r(x) = q - x$ preserves I. If $\alpha = -1$, then $r \circ h$ is a translation preserving I, so it is the identity and $h = r$. This is a nonidentity involution, excluded by odd group order. The primitive case, and the induction, are complete.

The affine lemma works even at $p = 3$. It also holds over every odd composite modulus when α is a unit, although the primitive case only needs a prime modulus.

Tournament corollary. A finite tournament has odd-order automorphism group. Otherwise Cauchy's theorem gives a nonidentity involution. Such a permutation swaps some distinct u, v, reversing the unique arc joining them, and therefore cannot be an automorphism. A cyclic tournament has τ among its automorphisms because translation preserves all differences. Apply the group theorem to $\mathrm{Aut}(T)$. Preserving the two colors is precisely preserving I_{2q+1} setwise, so only the identity can do so. This proves the conjecture for every q and every connector set.

4. The supplied proof and computational verification

The uploaded second-moment argument is also valid. For $p \geq 5$, put $m = (p+1)/2$ and work in $\mathbb{F}_p$. The centroid of I_p is $c = (m-1)/2 = -1/4$. The sum-of-squares identity gives

$$M = \sum_{j=0}^{m-1} (j-c)^2 = \frac{m(m^2-1)}{12} = -\frac{1}{32} \neq 0.$$

All denominators are invertible for $p \geq 5$. A preserving affine map fixes c and multiplies M by α^2, so $\alpha^2 = 1$. The positive multiplier gives the identity; the negative multiplier gives an involution. The supplied proof treats $p = 3$ separately, avoiding division by 3 there.

Actual graphs. The Python module constructs any input $T(n; S)$, validates S, and exports arcs, adjacency matrices, directed DOT files, and SVG diagrams. The included examples are:

Example	n	S	$\lvert \mathrm{Aut}(T) \rvert$	Colored order
Directed triangle	3	$\{1\}$	3	1
Consecutive connectors	5	$\{1, 2\}$	5	1
Paley tournament	7	$\{1, 2, 4\}$	21	1
Three triangles in a cycle	9	$\{1, 3, 4, 7\}$	81	1

The final example is the imprimitive composition $C_3[C_3]$. Checking translations alone would miss many of its automorphisms. The implemented search uses arbitrary adjacency-preserving permutations, not an assumed affine family.

Checks completed.

- All 4,094 connector sets of orders $3, 5, \ldots, 23$, plus the singleton control, passed exact canonical-coloring checks.
- An independent all-permutations oracle matched the automorphism engine in 2,258 cases.
- All 143,346 affine permutations over the 50 odd moduli from 3 to 101 passed the affine interval lemma check.
- The supplied sanity program was rerun: 167 odd primes below 1,000, 2,401 factor pairs, and all relevant odd-order color-preserving permutations through degree 13 passed.

These finite checks support the formulation and code; they are *not premises* of the all-orders proof. The unrelated conjecture searches in the Round41 archive are not used and have not been re-certified here.

Provenance. The separately attached PDF and the PDF inside the archive have identical text after whitespace normalization. The project includes input hashes, an independent proof audit, the new affine-lemma check, and machine-readable finite-check results.

5. Formalization and publication

Lean status: complete. On 30 October 2026, the project built successfully on Alex Chan's macOS machine with Lean 4.35.0-rc3. The supplied build output includes the following final declarations:

- `AlbertsonCollins.burnsideSchurStatement_proved`;
- `AlbertsonCollins.canonicalHalfGroup_proved`;
- `AlbertsonCollins.tournament_half_regular`.

The group theorem has no unproved mathematical hypothesis beyond the conditions in its statement. The tournament theorem takes a tournament on `Fin (2*q+1)` and the hypothesis that its standard rotation preserves arcs. It concludes regularity of `halfSet (2*q+1)` for the full automorphism group. This is the prescribed cyclic-label half-coloring conclusion, not a statement about arbitrary tournaments without the cyclic symmetry assumption.

Dependency audit. The final reports list only `propext`, `Classical.choice`, and `Quot.sound` under `#print axioms`. No `sorryAx` or project-specific axiom appears in those reports. The formerly explicit Burnside–Schur hypothesis is discharged in `BurnsideSchurBridge.lean`. The relevant composite-degree and prime-degree classification proofs are included under `lean/CFSG/Permutation/BurnsideSchur`. Their upstream revision is `69cb972f587a` of AxiomMath/CFSG. The files retain their Apache-2.0 license and source provenance in `lean/CFSG/LICENSE` and `lean/CFSG/UPSTREAM.json`.

Reproduction. The completed proof is pinned at project commit `2802921`. Its toolchain is Lean 4.35.0-rc3 and its Mathlib revision is `b63f6e8a68d2`. From the repository's `lean` directory, run:

```
lake update
lake exe cache get Mathlib
lake build
lake env lean AlbertsonCollins/BurnsideSchurBridge.lean
```

The final command prints the theorem signatures and their axiom reports. The machine-readable verification record is `lean/verification_status.json`. The build evidence described here is the successful output supplied by Alex Chan; the documentation update does not claim an additional independent Lean execution.

Development history. The 28 October package was an uncompiled foundation. The 29 October checkpoints established the block induction and coordinate transport. The completed 30 October version adds the interval and affine arguments, imports the audited classification proof, and closes the remaining hypothesis. Initial formalization notes and the original model audit are retained as dated historical records.

Paper status. The accompanying article now records the completed formalization and its external source attribution. It remains a submission draft; neither formal verification nor this review establishes priority, journal acceptance, or external human peer review. The earlier literature search checked primary papers and targeted later work, but did not exhaust MathSciNet, zbMATH, or unpublished work. That search has not been repeated as part of this documentation update. Journal-specific preparation notes are included separately.

Primary references checked.

1. K. Meslem and E. Sopena, *On the distinguishing number of cyclic tournaments: Towards the Albertson–Collins Conjecture*, Discrete Applied Mathematics **266** (2019), 219–236. Conjecture 2; arXiv:1608.04866v4.

2. M. O. Albertson and K. L. Collins, *A note on breaking the symmetries of tournaments*, Congressus Numerantium **136** (1999), 129–131. Conjecture 2.4; coauthor's primary copy.

3. S. Evdokimov and I. Ponomarenko, *A new look at the Burnside–Schur theorem*, Bulletin of the London Mathematical Society **37** (2005), no. 4, 535–546. Opening theorem; arXiv:math/0310373.

4. D. Gluck, *Trivial set-stabilizers in finite permutation groups*, Canadian Journal of Mathematics **35** (1983), no. 1, 59–67. Corollary 1; doi:10.4153/CJM-1983-005-2.

Continue from this package. Reproduce the completed proof using the pinned dependencies above. Publication preparation now concerns author review, journal declarations, and the literature and priority assessment, rather than an outstanding Lean proof obligation.